

BUGFIXES

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Root-cause-analysed bug fixes for this project, per the Universal Mandatory Constraints (CLAUDE.md mandate #10). Each entry records the defect, root cause, affected files, the fix, and a link to the reproduction/verification evidence.

BUGFIX-0001 — run-tests.sh aborts after the first test under set -e

- **Type:** Bug (script-internal failure — Helix Constitution §11.4.1)
- **Status:** Fixed
- **Date:** 2026-06-30
- **Affected file:** tests/run-tests.sh (test_result())

Symptom

bash tests/run-tests.sh printed only its banner and the first test's section header, then exited 1 (7 lines of output total) — every test after the first was silently never run, and no TEST SUMMARY was printed.

Reproduction (captured, run in-session before the fix)

```
$ bash -c 'set -euo pipefail; x=0; ((x++)); echo "SURVIVED,
x=$x"; echo "exit=$?"
exit=1                                # "SURVIVED" never printed

$ bash tests/run-tests.sh 2>&1 | wc -l
7                                     # aborted; exit=1
```

Root cause (FACT — not a guess, Helix Constitution §11.4.6)

`test_result()` counted with the bash post-increment idiom `(( TESTS_RUN++ ))`. The arithmetic command `(( expr ))` returns exit status **1** when `expr` evaluates to **0**. `TESTS_RUN++` is a *post*-increment, so its value is the value *before* incrementing — 0 on the very first call — making `(( TESTS_RUN++ ))` return status 1. The script runs under `set -euo pipefail` (line 7), so that non-zero status aborted the entire suite at the first `test_result` call, before any results or the summary could print. The same trap applied to the first PASS `(( TESTS_PASSED++ ))` and first FAIL `(( TESTS_FAILED++ ))`.

This was a **pre-existing** latent defect (the `(( ... ++ ))` idiom was original); it surfaced while wiring the constitution-inheritance pre-flight gate in as the first test.

Fix (at source, Helix Constitution §11.4.1)

Replaced the three post-increment arithmetic commands with the assignment form, which always returns status 0 and is immune to the `set -e` trap:

```
- ((TESTS_RUN++))
+ TESTS_RUN=$((TESTS_RUN + 1))
- ((TESTS_PASSED++))
+ TESTS_PASSED=$((TESTS_PASSED + 1))
- ((TESTS_FAILED++))
+ TESTS_FAILED=$((TESTS_FAILED + 1))
```

Verification (captured, run in-session after the fix)

```
$ bash -c 'set -euo pipefail; x=0; x=$((x+1)); echo "SURVIVED,
x=$x"; echo "exit=$?"
SURVIVED, x=1
exit=0
```

```
$ bash tests/run-tests.sh 2>&1 | wc -l
28                                # suite now runs to completion (was 7)
# first line of results:
✓ PASS: Constitution inheritance (§11.4.35)
```

The suite now executes every test and prints its summary. (Its overall exit remains non-zero in an environment without the running proxy services — those are real-infrastructure tests that correctly FAIL/skip when the live System is absent, Helix Constitution §11.4.11 — that is a separate, pre-existing, infrastructure-dependent condition, not this defect.)

Cross-reference (§11.4.138): the “proxy services not running” condition noted above was treated as merely environmental. It was not — BUGFIX-0002 below is the *reason* the proxy would not stay up under rootless Podman. Driving the real data path (not just “tests pass”) surfaced it.

BUGFIX-0002 — squid crash-loops under rootless Podman (log dir not writable) → proxy never serves

- **Type:** Bug (product defect — end-user-visible: the proxy does not work)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** lib/container-runtime.sh (create_directories()), start (init_cache())
- **Regression guard:** tests/regression/log_dir_writable_test.sh (§11.4.135, §11.4.115 RED_MODE polarity)
- **Workable item:** #38 (the live-usability proof stream)

Symptom

A freshly-booted proxy (./start --no-vpn, rootless Podman) returned http_code=000 for every request through it. proxy-squid was not (healthy) — it was restarting in a loop, never accepting a connection, so **no feature of the proxy worked for the end user** — the exact “tests pass but the feature can’t be used” failure mode §11.4 forbids.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Live boot of the real orchestrator, then curl through the proxy:
$ curl -s -o /dev/null -w '%{http_code}' -x http://127.0.0.1:53128 http://example.com/
000
```

```
# squid's own log (read from inside the container) showed the FATAL:
$ podman logs proxy-squid | tail
FATAL: Cannot open '/var/log/squid/access.log' for writing.
       The parent directory must be writeable by the user 'proxy',
       which is the cache_effective_user set in squid.conf.
```

```
# Unit reproduction of the orchestrator code path (pre-fix replica):
$ RED_MODE=1 tests/regression/log_dir_writable_test.sh
```

```
[PASS] BUGFIX#38 log-dir-writable (RED_MODE=1): RED reproduced:
pre-fix LOG_DIR mode=755 is NOT world-writable
```

Root cause (FACT — not a guess, §11.4.6)

Under **rootless** Podman the invoking host uid (1000) maps to the container's root, but every other container uid is remapped through `/etc/subuid` into a high host range. `squid.conf` sets `cache_effective_user proxy`, so squid drops privileges to its non-root proxy user — which, on the host, is a high subuid that owns **none** of the host-created bind-mount dirs. The host `./logs` directory was created mode `0755` (owner-write only), so the container proxy user had only world `r-x` — no write — and squid `FATALed` opening `access.log`, then crash-looped. The orchestrator already `chmod 777`-ed the **cache** bind-mount (`init_cache`) for this exact reason, but the **log** bind-mount was missed — a single missing site of an already-known remedy.

Fix (at source, §11.4.1 / §11.4.108 SOURCE layer)

Make the squid log bind-mount world-writable in the orchestrator's `dir-init`, mirroring the existing `cache chmod 777`. Applied at **both** self-sufficient sites so it holds regardless of call order:

```
# lib/container-runtime.sh  create_directories()
    mkdir -p "$LOG_DIR"
+  chmod 777 "$LOG_DIR"

# start  init_cache()
+  mkdir -p "$LOG_DIR"
+  chmod 777 "$LOG_DIR"
```

Verification (captured, run in-session after the fix)

Unit — regression guard, §11.4.115 polarity (RED reproduces, GREEN proves):

```
$ RED_MODE=1 tests/regression/log_dir_writable_test.sh  # pre-fix replica
[PASS] RED reproduced: LOG_DIR mode=755 NOT world-writable
$ tests/regression/log_dir_writable_test.sh              # real fixed code
[PASS] GREEN: create_directories() made LOG_DIR mode=777 world-writable
```

§1.1 paired mutation (guard is not a tautology): stripping the `chmod 777 "$LOG_DIR"` from `create_directories()` made the GREEN guard **FAIL** (REGRESSION: ... mode=755 NOT world-writable, exit 1); the

lib was restored byte-identical (md5
0128a96b6d467c2da5b7cef8a808e563 before == after) and the guard
PASSED again.

Live data-plane (§11.4.108 RUNTIME + USER-VISIBLE):

after the fix the booted proxy serves real HTTP through it:

```
$ curl -x http://127.0.0.1:53128 http://example.com/ ->  
http_code=200  
Via: 1.1 proxy-squid (squid/6.13)  
# squid access.log (the file that FATAL'd pre-fix) – real request  
logged:  
1782858066.886 189 10.89.1.249 TCP_MISS/200 951 GET http://  
example.com/ - HIER_DIRECT/104.20.23.154 text/html  
$ podman ps -> proxy-squid Up (healthy)
```

Evidence: qa-results/regression/bugfix38/.

BUGFIX-0003 — tests/run-tests.sh aborts mid-suite on a no-message FAIL (set -e); ~29 of 39 tests silently never ran

- **Type:** Bug (script-internal failure — §11.4.1; sibling of BUGFIX-0001)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected file:** tests/run-tests.sh (test_result())
- **Regression guard:** tests/regression/test_result_returns_zero_test.sh (§11.4.135, §11.4.115 RED_MODE polarity)

Symptom

bash tests/run-tests.sh printed only ~10 results then stopped — no TEST SUMMARY, exit 1. The Directory / Config / Compose / Runtime / Port / Cache / VPN / Service-startup groups after the first FAIL never ran, so the suite *appeared* to run while actually exercising under a third of its tests and hiding real failures.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Extract the pre-fix test_result and call it with a no-message  
FAIL under set -e:  
$ bash -c 'set -euo pipefail; <test_result>; test_result "x"  
"FAIL"; echo SURVIVED'  
x FAIL: x
```

```
# exit=1 – "SURVIVED" never printed (the function returned 1 and
aborted)
```

```
$ RED_MODE=1 tests/regression/test_result_returns_zero_test.sh
[PASS] RED reproduced: pre-fix test_result aborts under set -e on
a no-message FAIL
```

Root cause (FACT — not a guess, §11.4.6)

test_result()'s FAIL branch ended on `[[ -n "$message" ]] && echo -e " → $message"`. When message is empty (every test_result "... " "FAIL" call with no third argument), the `[[ -n "$message" ]]` test is false, the `&&` list short-circuits, and the list — the **last command of the function** — returns exit status **1**. When such a test_result is the last command executed in a test function (e.g. the final "Upstreams" iteration of test_directories's for loop, where the dir is absent → FAIL), that function returns 1, and under set -euo pipefail main() aborts immediately — every later test group and the summary never run. This is the same §11.4.1 class as BUGFIX-0001 (a reporting helper leaking a non-zero status into control flow), a *different* site (test_result's tail, not the ((++)) counter), so BUGFIX-0001's fix did not cover it.

Fix (at source, §11.4.1)

test_result is a reporting helper; its exit status must never gate control flow. Append an explicit return 0:

```
        [[ -n "$message" ]] && echo -e "  ${YELLOW}→ $message$
        {NC}"
    fi
+
+   return 0
+ }
```

Verification (captured, run in-session after the fix)

```
# §11.4.115 GREEN (real fixed test_result survives a no-message
FAIL):
```

```
$ tests/regression/test_result_returns_zero_test.sh
[PASS] GREEN: real test_result returns 0 on a no-message FAIL
```

```
# Full suite now runs to completion (was 32 lines, aborted):
```

```
$ bash tests/run-tests.sh  # 82 lines, reaches TEST SUMMARY
Tests Run:    41
Tests Passed: 34
Tests Failed: 7
```

§1.1 paired mutation (guard is not a tautology): deleting the return 0 re-introduced the abort → the GREEN guard **FAILED** (REGRESSION: test_result aborts the suite ..., exit 1); tests/run-tests.sh was restored byte-identical (md5 7c2bab18c4566d081b1c8aa7a9a412e0 before == after) and the guard PASSED again.

Follow-up (separate, tracked): with the abort fixed, the suite now honestly reports **7 pre-existing FAILs** (e.g. Directory Upstreams) that the abort had been masking. These are real existing-system findings to triage — not regressions from this fix, but newly *visible* because the suite finally runs to completion.

Evidence: qa-results/regression/bugfix0003/.

BUGFIX-0004 — run-tests.sh reports FAILURE on a HEALTHY serving proxy (§11.4.1 false-FAIL); skips recorded as PASS

- **Type:** Bug (anti-bluff — §11.4.1 false-FAIL + §11.4.3 PASS-by-default-for-skip)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected file:** tests/run-tests.sh (test_result, test_environment, test_directories, test_container_runtime, test_ports + new port_verdict/ports_check_one, test_vpn, test_service_startup, print_summary)
- **Regression guard:** tests/regression/port_topology_aware_test.sh (§11.4.135, §11.4.115 RED_MODE polarity)
- **Bluff audit:** docs/research/existing_test_bluffs_audit/README.md (B7)
- **Workable item:** #47 (P8)

Symptom

Once BUGFIX-0003 let the suite run to completion, bash tests/run-tests.sh reported Tests Run: 41, Passed: 34, Failed: 7 and exited **1 against a perfectly healthy, serving proxy** (proxy-squid + proxy-dante Up (healthy)). A suite that declares FAILURE on a working System is a §11.4.1 false-FAIL — as misleading as a false-PASS.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Against the running, healthy proxy:
$ bash tests/run-tests.sh | tail -4
Tests Run:      41
Tests Passed: 34
Tests Failed: 7      # exit 1
# The 7 FAILs: .env file exists, HTTP_PROXY_PORT set, Directory
Upstreams,
# Docker installed, Port 53128 available, Port 51080 available,
Port 58080 available

$ RED_MODE=1 tests/regression/port_topology_aware_test.sh
[PASS] RED reproduced: pre-fix logic classifies a healthy serving
proxy port
      (owner up + listening) as FAIL
```

Root cause (FACT — not a guess, §11.4.6)

Three independent bluff classes, all FACT-confirmed against the live host:

1. **Port-semantics inversion (§11.4.1).** `test_ports` reported any port that was IN USE as FAIL (“Port in use”). The running proxy’s own listening ports (squid 53128, dante 51080) are *in use by their own healthy service* — the serving state — so the suite FAILED on health.
2. **Inverted runtime check (§11.4.161).** `test_container_runtime` asserted Docker installed and FAILED on its absence. The project MANDATES rootless Podman and FORBIDS Docker as a workflow — the check tested for the forbidden runtime and failed on the compliant state.
3. **Config-absent-as-FAIL (§11.4.3).** `.env` (gitignored, §11.4.10/ §11.4.30) and `Upstreams/` absence FAILED, though their absence is the *expected* fresh-checkout topology. And (B7) `test_result` had no SKIP state, so `test_vpn/test_service_startup` skipped paths were recorded as PASS, inflating `TESTS_PASSED` (§11.4.3 PASS-by-default).

Fix (at source, §11.4.1 / §11.4.3 / §11.4.161)

- **3-state test_result** — add a SKIP branch + `TESTS_SKIPPED` counter + Tests Skipped: summary line; the suite exit gates on `TESTS_FAILED -eq 0` only (skips are neither pass nor fail). BUGFIX-0001 assignment-form counters + BUGFIX-0003 return 0 preserved.
- **§11.4.3 topology-aware ports** — a *pure* `port_verdict(owner_serving, listening)` truth-table (serving→PASS, owner-up-but-not-listening→FAIL, pre-start-

free→PASS, foreign-process-holds-it→SKIP); _ports_check_one resolves owner_serving from real podman ps/podman port and listening from real ss (no data-plane probe).

- **§11.4.161** — assert the mandated podman; Docker informational only, absence is SKIP not FAIL.
- **§11.4.3 SKIP** for absent gitignored .env/HTTP_PROXY_PORT (validate the tracked .env.example template instead) and absent upstreams//Upstreams/.
- **B7** — test_vpn/test_service_startup disabled paths emit SKIP.

Verification (captured, run in-session after the fix)

Full suite vs the healthy running proxy – zero failures, honest skips:

```
$ bash tests/run-tests.sh | tail -5
```

```
Tests Run:      43
```

```
Tests Passed:   37
```

```
Tests Skipped:  6
```

```
Tests Failed:   0          # exit 0 – "All tests passed (6 skipped ...)"
```

§11.4.115 polarity guard (tests the REAL pure port_verdict, not a copy):

```
$ RED_MODE=1 tests/regression/port_topology_aware_test.sh  #  
reproduces
```

```
[PASS] RED reproduced: healthy serving port classified FAIL
```

```
$ tests/regression/port_topology_aware_test.sh              #  
GREEN
```

```
[PASS] GREEN: HEALTHY=PASS NOTSERVING=FAIL PRESTART_FREE=PASS  
PRESTART_BUSY=SKIP
```

§1.1 paired mutation (guard is not a tautology): flipping port_verdict's PASS/FAIL branch made the GREEN guard **FAIL** (REGRESSION: ... HEALTHY=FAIL ..., exit 1) with bash -n still clean (assertion, not parse error); tests/run-tests.sh restored byte-identical (md5 2f4d3e4bf33cd391036cee595839d439) and the guard PASSED again. The guard is wired into test_regression_guards() (GREEN + RED self-check); the 5 pre-existing guards still PASS.

Honest note (§11.4.6 / §11.4.174): port 58080 (the control-API port) is currently held by a **non-project** host process — no proxy container publishes it — so the suite classifies it SKIP (readiness not assertable), never a false-FAIL blaming the proxy. The foreign process was **not** touched (shared-host ownership). This is a real condition to free before P10 deploys the control-API there.

Evidence: qa-results/p8/, qa-results/regression/portstopology/.